

KAPUSKASING, ON, Canada

WMO: 712990

Lat: 49.4053N

Lon: 82.4431W

Elev: 715

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.7	-24.4	-37.0	0.7	-29.4	-31.6	0.9	-24.1	21.4	9.6	19.8	10.4	3.7	290	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	84.2	67.5	81.0	65.6	77.8	64.1	70.4	79.8	68.4	76.9	66.5	74.5	11.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	103.0	74.6	65.3	95.9	73.0	63.3	89.3	71.1	34.9	79.8	33.1	76.9	31.6	74.7	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	18.5	16.7	DB	-37.7	90.4	5.7	3.5	-41.9	92.9	-45.2	94.9	-48.4	96.9	-52.6	99.4	(4)
(5)			WB	-37.9	74.2	5.7	1.9	-42.0	75.5	-45.4	76.6	-48.6	77.7	-52.7	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	0.6	4.6	17.5	32.8	48.3	59.1	63.7	61.3	53.4	40.3	25.7	10.4		
	DBStd	24.47	14.90	13.44	14.11	10.53	9.51	8.27	6.54	6.81	8.63	8.92	11.51	13.97		
	HDD50	6850	1532	1272	1009	520	149	17	1	3	59	328	731	1229		
	HDD65	11122	1997	1692	1472	965	524	211	106	155	360	766	1180	1694		
	CDD50	1365	0	0	2	5	97	291	424	354	163	28	1	0		
	CDD65	159	0	0	0	0	7	34	64	41	13	0	0	0		
	CDH74	1657	0	0	1	3	126	463	601	361	99	3	0	0		
	CDH80	386	0	0	0	0	33	123	140	73	17	0	0	0		
	WSAvg	7.9	7.8	8.2	8.4	8.4	8.2	7.3	7.1	7.2	7.7	8.4	8.6	8.1		
	PrecAvg	33.5	1.7	1.6	2.2	2.2	3.0	3.5	4.0	3.8	3.7	3.2	2.9	2.0		
	PrecMax	45.1	4.7	3.3	5.1	5.2	7.1	7.1	8.3	8.3	6.2	7.4	6.1	3.3		
(15) Precipitation	PrecMin	24.7	0.4	0.5	0.7	0.4	0.4	1.7	0.9	1.7	0.9	1.1	1.1	1.3		
	PrecStd	5.3	0.8	0.7	1.1	1.1	1.5	1.4	1.6	1.7	1.4	1.4	1.2	0.6		
	DB	36.6	42.1	58.9	69.0	84.5	88.8	88.6	86.1	82.2	71.5	56.6	40.1			
	MCWB	35.2	37.6	48.7	51.9	65.1	68.1	69.7	68.9	67.3	60.9	50.5	37.4			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	33.2	35.4	48.5	60.9	77.0	83.3	83.7	81.7	76.6	65.1	49.4	36.0			
	2% MCWB	32.2	32.5	41.1	46.6	59.6	65.4	67.9	67.3	65.4	56.7	45.4	34.3			
	5% DB	30.0	30.7	42.5	54.7	71.7	79.4	80.3	78.1	72.6	59.7	44.5	33.4			
	5% MCWB	28.7	28.4	36.6	43.5	56.4	62.7	66.2	65.7	63.3	53.8	41.0	32.3			
	10% DB	24.5	26.0	37.6	48.9	66.4	75.3	77.1	74.4	68.2	54.5	40.4	30.7			
	10% MCWB	23.3	24.2	33.2	40.0	53.6	61.2	64.6	63.6	60.8	50.0	38.1	29.7			
	0.4% WB	35.4	38.4	51.0	54.1	68.8	72.1	73.3	72.2	70.4	62.4	51.6	38.1			
	0.4% MCDB	36.6	41.6	56.4	62.7	79.4	85.4	82.4	81.4	77.7	68.7	54.8	39.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	32.5	32.6	41.4	48.4	62.1	68.6	70.4	69.5	67.0	58.2	46.2	34.6			
	2% MCDB	33.2	35.2	47.0	59.5	73.2	78.6	80.0	77.3	73.8	63.4	49.0	35.7			
	5% WB	28.7	28.5	36.8	44.5	58.1	65.7	68.4	67.4	64.5	54.4	41.4	32.3			
	5% MCDB	29.9	30.6	41.6	53.2	67.7	74.3	76.9	75.7	70.8	59.1	43.9	33.3			
	10% WB	23.4	24.3	33.4	40.9	55.1	63.0	66.4	65.2	61.5	49.9	38.1	29.6			
	10% MCDB	24.5	26.1	37.5	48.2	64.5	71.8	74.0	73.0	67.7	54.6	40.7	30.7			
	MDBR	20.1	23.0	23.6	21.5	23.4	23.0	21.4	20.5	18.8	14.3	12.7	15.9			
	5% DB	16.6	21.0	24.9	30.4	32.4	29.1	25.5	25.2	24.1	22.5	16.9	14.2			
(36) Mean Daily Temperature Range	MCWBR	16.3	18.7	17.9	19.0	17.9	15.0	13.2	13.3	14.4	15.4	13.8	13.5			
	MCDBR	16.5	20.2	21.8	26.5	26.9	22.6	21.3	21.2	21.2	20.5	15.6	13.8			
	MCWBR	16.3	18.1	16.4	17.6	16.8	13.7	12.7	12.6	14.3	15.3	13.5	13.4			
	MCDBR	16.3	18.1	16.4	17.6	16.8	13.7	12.7	12.6	14.3	15.3	13.5	13.4			
(40) Clear-Sky Solar Irradiance	taub	0.268	0.266	0.299	0.339	0.361	0.372	0.395	0.374	0.365	0.319	0.282	0.254			
	taud	2.251	2.279	2.268	2.327	2.404	2.393	2.352	2.410	2.425	2.535	2.500	2.353			
	Ebn at Noon	248	282	289	288	285	282	272	274	264	261	248	241			
	Edh at Noon	23	28	35	36	36	36	37	34	30	23	18	18			
(44) All-Sky Solar Radiation	RadAvg	389	737	1186	1494	1643	1773	1724	1441	1046	613	328	261			
	RadStd	58	71	94	121	131	144	170	98	92	80	32	39			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page